

RDK S600 board introduction

RDK S600 board introduction

1. Preparations Before Use
2. Overview
 - 2.1 **Parameter**
 - 2.2 Architecture Diagram
 - 2.3 Interface Diagram
 - 2.4 Main Functions

1. Preparations Before Use

1. Power Supply

The RDK S600 development board is powered via a DC connector. It is recommended to use the official power adapter included in the kit.

2. Storage

The RDK S600 uses eMMC as the system boot media, providing sufficient storage space for the Ubuntu system and application software.

3. Display

The RDK S600 development board features an HDMI display interface. Connect the board to a monitor via an HDMI cable for a graphical desktop display.

4. Network Connection

The RDK S600 development board supports both Ethernet and Wi-Fi. Users can connect to the network via either interface.

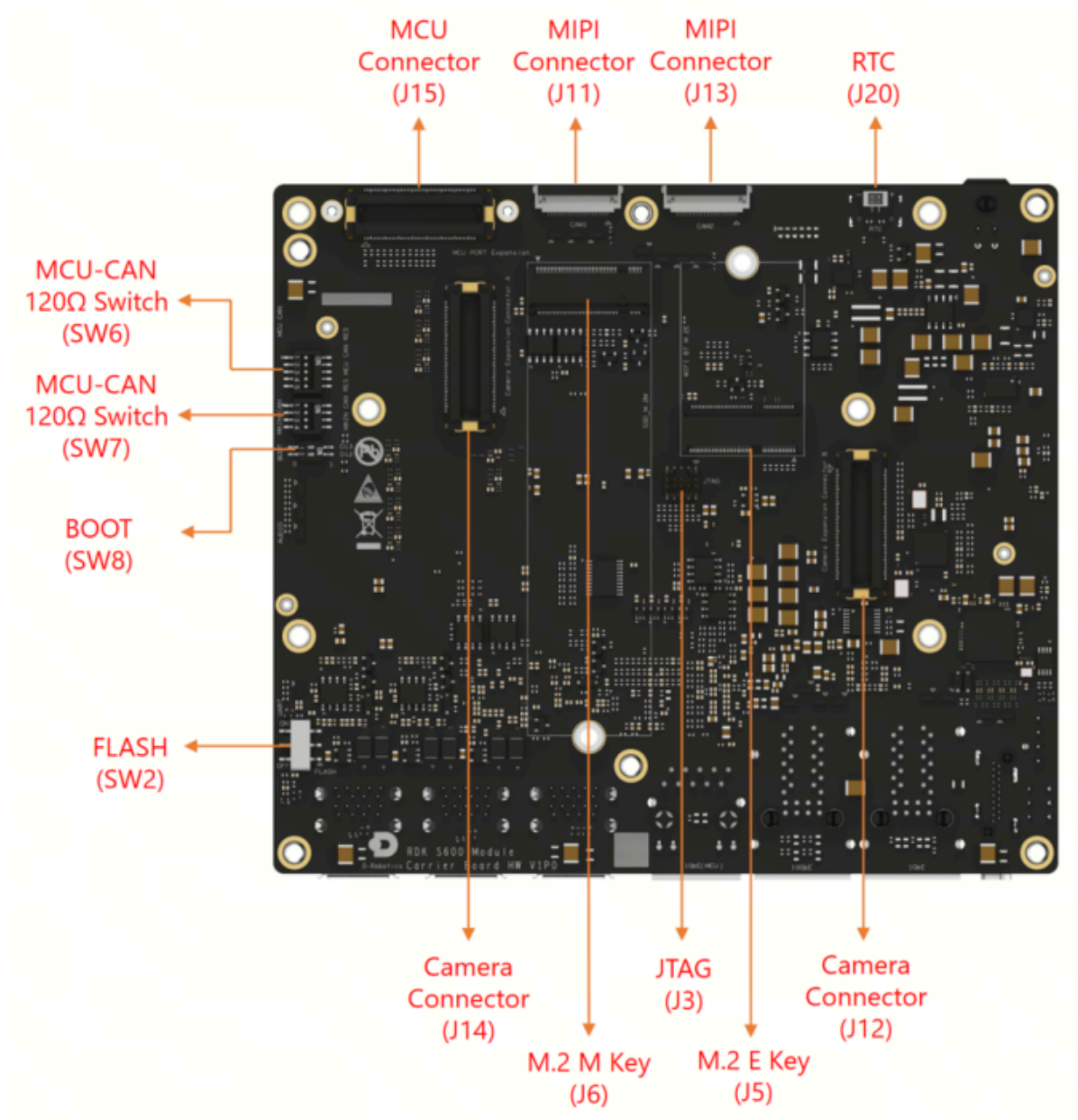
2. Overview

The D-Robotics RDK S600 Series Developer Kit is equipped with the S600 intelligent computing chip, with the BPU providing up to 560 TOPS of computing power. It is a development board designed for intelligent computing and robotics applications, featuring rich interfaces and extreme ease of use. Its unique heterogeneous design can simultaneously handle perception inference and real-time motion control needs, reducing the volume and complexity of the control system.

2.1 Parameter

Category	Specifications
CPU	• 18x Arm® Cortex®-A78AE CPU 2.0GHz
MCU	• 6x Arm® Cortex®-R52+ MCU (1× DCLS, 2× Split-Lock)
BPU	• 4x BPU Nash core provides up to 560 TOPS
Memory	• 32/64GB LPDDR5, 256-bit, up to 6400MT/s
Storage	• 64/256GB UFS 3.1 • M.2 Key M Connector for NVMe SSD
USB	• 6x USB 3.2 Gen 1x1 Type-A • 1x USB 2.0 Type-C (for flashing and debugging)
Network	• M.2 Key E Connector for Wi-Fi&BT Module • 2x 1GbE RJ45 Ports • 2x 10GbE RJ45 Ports • 1x 1GbE RJ45 Ports (MCU-Domain)
Display	• 1x HDMI
Expansion	• 2x Camera Expansion Connector • 1x MCU Port Expansion Connector
Connectors	• 1x 2-pin RTC Battery Connector • 1x 4-pin Fan Connector • 1x 12-pin Automotive Connector • 1x JTAG Debug Connector (for Main and MCU) • 1x 12-pin MCU-Domain CAN(5x) Connector • 1x 10-pin Main-Domain CAN(4x) Connector • 1x 10-pin MCU-Domain UART(2x) & Main-Domain UART(2x) Connector • 1x 10-pin PCM Connector • 2x 22-pin MIPI Connector
Power	• 12~28V • 4-pin Connector
Size	• 140mm x 123mm x 78mm

2.2 Architecture Diagram



2.4 Main Functions

No.	Function	No.	Function	No.	Function
D59	System LED	J11	MIPI Camera Interface	J22	Main Board Function Interface
D60	Power LED	J12	Camera Expansion Interface	K1	RST Button
D61	Flash LED	J13	MIPI Camera Interface	K2	WAKE Button
J1	Main Board Power Interface	J14	Camera Expansion Interface	SW2	Flash Switch
J2	699-Pin B2B Connector	J15	MCU Expansion Interface	SW3	Power Switch
J3	JTAG Interface (MCU&MAIN)	J16	MCU-CAN Interface	SW6	MCU-CAN Resistor Selection
J4	Flash (Flashing, Main&MCU Debug)	J17	MAIN-CAN Interface	SW7	MAIN-CAN Resistor Selection
J5	M.2 Key E Interface	J18	UART Interface (MAIN&MCU)	SW8	BOOT
J6	M.2 Key M Interface	J19	PCM+I2C Interface	U44	2x 1GbE
J7/J8/J9	USB 3.0 Type-A Interface	J20	RTC Battery Interface	U45	2x 10GbE
J10	HDMI Interface	J21	Fan Interface	U80	1x 1GbE (MCU)